1. Introduction:

In the realm of global health, hypertension is one of the most common and significant risk factors for cardiovascular diseases, stroke, and other chronic conditions. With the increasing prevalence of hypertension worldwide, understanding its distribution across countries, regions, and demographics is crucial for designing targeted healthcare interventions and policies. The **World Bank Hypertension Prevalence Dataset** provides a comprehensive overview of hypertension rates in adults across various countries and regions over multiple years, allowing researchers, policymakers, and public health experts to analyze trends and disparities in hypertension prevalence.

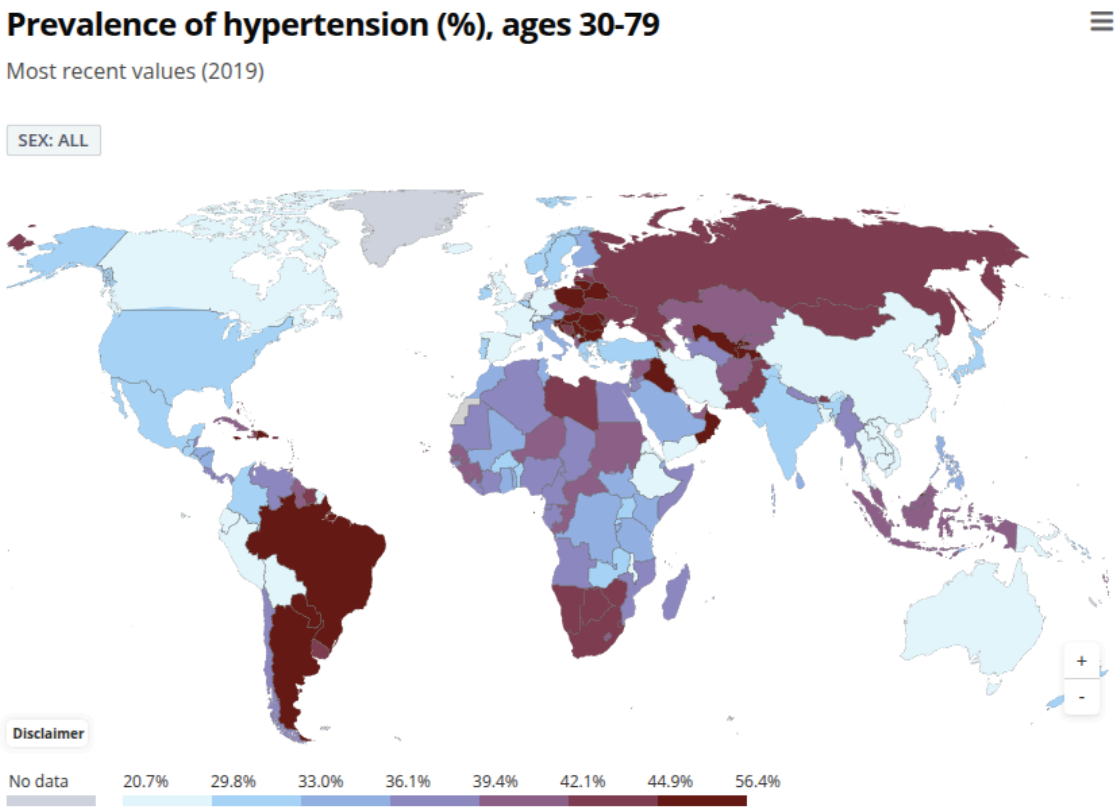
This dataset offers valuable insights into how hypertension varies by factors such as geography, age, sex, and urbanization. In this report, our goal is to demonstrate the **data preprocessing steps** involved in preparing this real-world dataset for analysis. We will cover key techniques such as data cleaning, feature scaling, encoding categorical variables, data splitting, and other essential steps that ensure the dataset is ready for further modeling or statistical analysis. By walking through these preprocessing stages, we aim to provide a clear understanding of how to prepare the dataset for machine learning or other analytical methods, and how these steps can help in drawing meaningful insights from the data.

2. Dataset Documentation:

2.1 Dataset Overview

The World Bank Hypertension Prevalence Dataset focuses on the prevalence of hypertension among adults aged 30-79 years. The data is collected from a range of 192 economies globally, spanning the years 1990 to 2019. The dataset is updated annually and provides age-standardized statistics on the percentage of adults with hypertension, defined as having systolic blood pressure ≥140 mmHg, diastolic blood pressure ≥90 mmHg, or taking medication for hypertension.

- Indicator: Prevalence of hypertension (%), ages 30-79.
- Time Period: Data from 1990 to 2019.
- Geographical Coverage: 192 economies worldwide.
- Frequency: The dataset is updated annually and was last updated 11 days ago.
- Source: Global Health Observatory (GHO), World Health Organization (WHO).
- License: The dataset is provided under the Creative Commons 4.0 Attribution International License (CC BY 4.0).



The data in this dataset is age-standardized, meaning it accounts for age differences between populations to allow for a more accurate comparison. Hypertension is diagnosed based on the criteria mentioned above, and this indicator is part of the Human Capital Project (HCP) under the World Bank's Health Indicators initiative.

2.2 What the Dataset is About

The dataset provides information on the prevalence of hypertension across multiple countries and regions, broken down by age group and sex. Here is a summary of the key details:

- Rows: The dataset contains 17,280 rows (each representing a unique combination of country/region, year, and specific demographic factors).

- Columns: There are 39 columns, including key identifiers such as STRUCTURE_ID, REF_AREA_LABEL, SEX_LABEL, AGE_LABEL, and the OBS_VALUE (the observed hypertension prevalence).

The dataset provides a time-series view of hypertension rates from 1990 to 2019, making it possible to track changes in hypertension prevalence over time across different countries, regions, and demographic groups (e.g., sex, age, urbanization).

Columns

The dataset contains **39 columns**, each representing a specific aspect of the data. Below is a detailed description of the key columns:

1. **STRUCTURE**:
    - Refers to the structure of the data (e.g., the data type or aggregation level).
2. **STRUCTURE_ID**:
    - A unique identifier for the data structure.
3. **ACTION**:
    - Specifies the action type for the data (e.g., collection method or survey type).
4. **FREQ**:
    - The frequency at which the data is reported (e.g., annually).
5. **FREQ_LABEL**:
    - Describes the frequency of data collection (e.g., "Yearly").
6. **REF_AREA**:
    - The reference area or geographical location (e.g., country or region).
7. **REF_AREA_LABEL**:
    - The human-readable label for the reference area (e.g., "United States", "Africa").
8. **INDICATOR**:
    - A specific indicator or measurement related to hypertension prevalence.
9. **INDICATOR_LABEL**:
    - A detailed description of the indicator (e.g., "Prevalence of Hypertension").
10. **SEX**:
    - Gender breakdown of the data (e.g., Male, Female, or Both).

11. **SEX_LABEL**:
    - A descriptive label for the gender breakdown (e.g., "Male", "Female").
12. **AGE**:
    - The age group for which the data is available (e.g., "18+", "45-64", etc.).
13. **AGE_LABEL**:
    - The descriptive label for the age group.
14. **URBANISATION**:
    - Refers to whether the data is broken down by urban vs rural populations.
15. **URBANISATION_LABEL**:
    - A label for the urbanization category (e.g., "Urban", "Rural").
16. **UNIT_MEASURE**:
    - The unit of measurement for the data (e.g., percentage).
17. **UNIT_MEASURE_LABEL**:
    - A descriptive label for the unit of measurement (e.g., "Percentage").
18. **COMP_BREAKDOWN_1, 2, 3**:
    - Breakdown variables that categorize the data further, such as by socio-economic status, education level, or specific sub-regions.
19. **COMP_BREAKDOWN_LABEL_1, 2, 3**:
    - Descriptive labels for the breakdown categories.
20. **TIME_PERIOD**:
    - The time period for which the data is reported (e.g., a specific year or year range).
21. **OBS_VALUE**:
    - The observed value of hypertension prevalence (typically a percentage).

22. **DECIMALS**:
    - Number of decimal places in the observed value.
23. **DECIMALS_LABEL**:
    - A label for the number of decimal places used in reporting.
24. **DATABASE_ID**:
    - The identifier for the database from which the data is sourced.
25. **DATABASE_ID_LABEL**:
    - A label for the database (e.g., "World Health Organization").
26. **UNIT_MULT**:
    - A multiplier for the unit of measurement, if applicable.
27. **UNIT_MULT_LABEL**:
    - Describes the unit multiplier.
28. **UNIT_TYPE**:
    - The type of unit used for measurement (e.g., "Percentage").
29. **UNIT_TYPE_LABEL**:
    - A human-readable description of the unit type.
30. **TIME_FORMAT**:
    - The format in which time is recorded (e.g., "Annual").
31. **TIME_FORMAT_LABEL**:
    - A label describing the time format.
32. **OBS_STATUS**:
    - The status of the data observation (e.g., whether it's estimated or confirmed).
33. **OBS_STATUS_LABEL**:
    - A descriptive label for the observation status.

34. **OBS_CONF**:

    - The confidence level of the data observation.

35. **OBS_CONF_LABEL**:

    - A label describing the confidence level.

Rows

The dataset consists of **17,280 rows** representing individual **observations** of hypertension prevalence. Each row typically corresponds to a specific **country, region, or demographic breakdown** for a given **year**. This means that each row holds the data for a particular **country's hypertension prevalence** for a specific **time period** and possibly broken down by variables like **age**, **gender**, or **urbanization**.